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الجامعة العربية المفتوحة
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Faculty of Computer Studies

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T216A
Cisco Networking (CCCNA) part A

Midterm Examination (MTA)
Spring Semester 2025/2026

Date: 14/04/2026

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 2h

Instructions:

- Total Marks: **50**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Part 1: Multiple choices questions- 1 mark per question. **Total: 5 Marks**

1. How is SSH different from telnet?
   a. SSH makes connections over the network, whereas Telnet is for out-of-band access
   b. SSH provides security to remote sessions by encrypting messages and using user authentication. Telnet is considered insecure and sends messages in plaintext.
   c. SSH must be configured over an active network connection, whereas Telnet is used to connect to a device from a console connection.
   d. SSH requires the use of the PuTTY terminal program. Tera Term must be used to connect to devices through the use of Telnet.
2. After making configuration on a Cisco switch, a network administrator issues a ***copy running-config startup-config*** command. What is the result of issuing this command?
   a. The new configuration will be loaded if the switch is restarted.
   b. The new configuration will be stored in flash memory.
   c. The current IOS file will be replaced with the newly memory file.
   d. The configuration changes will b removed and the original configuration will be restored.
3. What layer is responsible for routing messages through an internetwork in the TCP/IP model?
   a. internet
   b. transport
   c. session
   d. network access
4. What is the purpose of the FCS fields in a frame?
   a. To obtain the MAC address of the sending node
   b. To verify the logical address of the sending node
   c. To compute the CRC header for the data field
   d. To determine if errors occurred in the transmission and reception
5. What OSI physical layer term describes the measure of the transfer of bits across a medium over a given period of time?
   a. Throughput
   b. Bandwidth
   c. Latency
   d. Goodput

Part 2: Answer all four questions – 4 marks per question. **Total: 16 Marks**

1. Answer the following questions:
   a. Briefly discuss the similarity and differences between **Store-and-forward switching** and **fragment-free switching** (2 marks)
   b. Briefly explain what do we mean by logical topology? (1 mark)
   c. What is meant by full-duplex communication? (1 marks)
2. Answer the following questions:
   a. What is the role the LLC sublayer? (1 mark)
   b. What is the maximum size of the Ethernet frame? (1 mark)
   c. What will a device do with a received Ethernet frame whose size is greater than the maximum allowed size? (1 mark)

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d. What will happen when a device receives an Ethernet frame, and the destination MAC address does not match the physical MAC address that is stores in RAM? (1 mark)

3. Answer the following questions:
   a. What is the min size of an IPv4 header (1 mark)
   b. The routing table stores three types of route entries; mention tow of these entries. (1 mark)
   c. Match the corresponding IPv6 fields to their proper description. (2 marks)
      Note: not all the fields will be used.
      IPv6 fields:
      - Payload length
      - Flow label
      - Hop limit
      - Source IPv6 address
      - Destination IPv6 address
      - Next header
      - Traffic class
      - Version

	Description	Field
A	This 20-bit field suggest that all packets with the same flow label receive the same type of handling by routers	
B	This field contains a 4-bit binary value set to 0110 that identifies this an IP version 6 packet	
C	This 8-bit field replaces the IPv4 TTL field	
D	This 16-bit field indicates the length of the data portion of the IPv6 packet.	

4. Answer the following questions:
   a. If a host didn't have a default gateway configured, what do you think would happen when it tries to send a packet to a device on another network? (1 mark)
   b. What is the purpose of using the ***service password-encryption*** command applied on a cisco router/switch? (1 marks)
   c. Why does a host use the ARP protocol? (1 mark)
   d. What happen when a device finds that there is no ARP table entry for a specific IPv4 address (1 mark)

Part 3: Answer all the questions **Total: 29 Marks**

1. Answer the following questions: (15 marks)
   a) Having the following prefix **145.50.0.0** and subnet mask **255.255.255.252**, determine: (7 marks)
      1. the prefix length (1.5 marks)
      2. the network address (1.5 marks)
      3. the first host address (1.5 marks)
      4. the last host address (1.25 marks)
      5. the broadcast address (1.25 marks)
   b) A PC has the following IP address **175.50.92.166** and subnet mask **255.255.255.240**. Determine the network address this PC belongs to. (8 marks)
      b1) Convert the dotted decimal IP address **175.50.92.166** to dotted binary; the steps of calculation in details need to be included for the last byte: **166.** (4 marks)
      Note1: 0.75 mark will be awarded for each correct byte converted
      Note2: 1 mark will be awarded for the details.
      b2) Convert the dotted decimal subnet mask 255.255.255.240 to dotted binary; (1 marks)

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b3) Determine the network address in dotted binary from the dotted binary IP address obtained in part **d1**, and the subnet mask obtained in part d2). Show the steps of calculation in details. (2 marks)
Note1: 1 mark for the details and 1 mark for the correct answer.

b4) From the dotted binary network address obtained in part d4) determine the network address in dotted decimal. (1 marks)

2. Answer the following questions (14 marks)

a) Write the command(s) needed to configure a switch with the following instructions: (6 marks)

- Set the switch's name to ***S15-F2-R3*** (1 mark)
- Secure the privilege EXEC mode with the most secure way using the following password ***Tar#27@*** (1 mark)
- Configure the console mode with the following password ***Dawn#3@*** (2 marks)
- Configure the interface VLAN1 with the following IP: 199.75.25.88 and subnet 255.255.255.240 (2 marks)

b) Convert the decimal number **231**, to binary, then to hexadecimal(2 marks)
Note: show the steps of calculation

c) Convert the hexadecimal number **B5**, to binary, then to decimal. (2 marks)
Note: show the steps of calculation

d) Refer to figures 1 and 2, and determined to which port(s) the switch (image 1) will forward the frame. (2 marks)

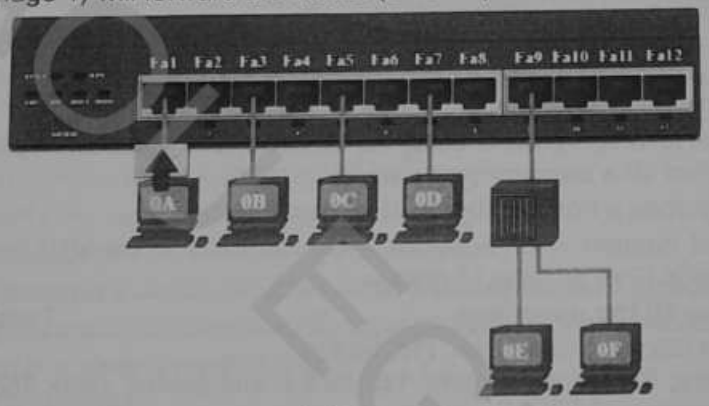


Image 1 Switch connection

Preamble	Destination MAC	Source MAC	Type / Length	Frame	End of Frame
	0B	0A			

Figure 1: Ethernet frame

Fa1	Fa2	Fa3	Fa4	Fa5	Fa6	Fa7	Fa8	Fa9	Fa10	Fa11	Fa12
0A						0D		0E 0F			

Figure 2: MAC address table

e) What type of cable should be used to connect two switches via Ethernet ports. (1 mark)

f) If a switch received a broadcast frame what will happen? (1 mark)

End of questions

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